903962578

ECE 2040 HW 3

Saahas Swaroop

Problem 4.2-2

Determine the node voltages for the circuit of Figure P 4.2-2.

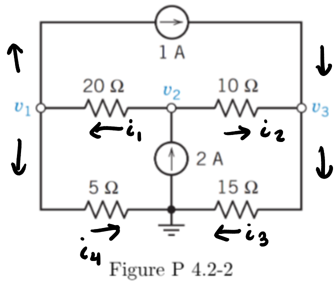

Figure P 4.2-2

$i_3 + i_4 = 2$

$i_2 + 1 = i_3$

$i_1 + i_2 = 2$

$i_1 = 1 + i_4$

$i_1 = \frac{v_2 - v_1}{20}$

$i_2 = \frac{v_2 - v_3}{10}$

$i_3 = \frac{v_3}{15}$

$i_4 = \frac{v_1}{5}$

$\frac{v_2 - v_1}{20} + \frac{v_2 - v_3}{10} = 2$

$v_2 - v_1 + 2v_2 - 2v_3 = 40$

$\rightarrow -v_1 + 3v_2 - 2v_3 = 40$

$\frac{v_2 - v_1}{20} = 1 + \frac{v_1}{5}$

$v_2 - v_1 = 20 + 4v_1$

$\rightarrow -5v_1 + v_2 = 20$

$\frac{v_3}{15} + \frac{v_1}{5} = 2$

$v_3 + 3v_1 = 30$

$\frac{v_2 - v_3}{10} + 1 = \frac{v_3}{15}$

$3v_2 - 3v_3 + 30 = 2v_3$

$\rightarrow 3v_2 - 5v_3 = -30$

$$V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad \begin{bmatrix} -1 & 3 & -2 \\ -5 & 1 & 0 \\ 0 & 3 & -5 \end{bmatrix} V = \begin{bmatrix} 40 \\ 20 \\ -30 \end{bmatrix}$$

Problem 4.2-7

The node voltages in the circuit shown in Figure P 4.2-7 are $v_a = 7\,V$ and $v_b = 10\,V$. Determine values of the current source current, i_s, and the resistance, R.

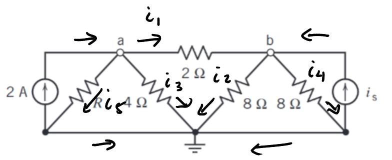

Figure P 4.2-7

$i_1 = \frac{7-10}{2} = -1.5A$

$i_2 = \frac{10}{8} = 1.25A$

$i_3 = \frac{7}{4} = 1.75A$

$i_4 = i_2 = 1.25A$

$i_5 = i_3 = 1.75A$

$i_s + i_1 = i_2 + i_4$

$i_s - 1.5 = 2.5$

$i_s = 4A$

$i_5 = \frac{7}{R}$

$1.75 = \frac{7}{R}$

$R = 4\Omega$

Problem 4.3-2

The voltages v_a, v_b, v_c, and v_d in Figure P 4.3-2 are the node voltages corresponding to nodes a, b, c, and d. The current i is the current in a short circuit connected between nodes b and c. Determine the values of v_a, v_b, v_c, v_d and of i.

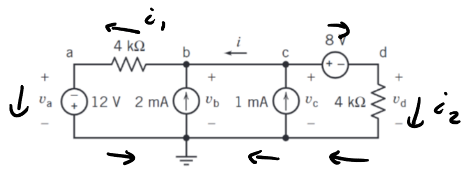

Figure P 4.3-2

Supernodes:

a & ground

c & d

$\frac{V_b - V_a}{4k\Omega} = i_1$

$\frac{V_d}{4k\Omega} = i_2$

$V_c - V_d = 8$

$0 - V_a = 12$

$V_a = -12\,V$

$i_1 = i + 2mA$

$i_2 + i = 1mA$

$i = 2mA - i_1$

$i = 1mA - i_2$

$\rightarrow 2mA - i_1 = 1mA - i_2$

$1mA = i_1 - i_2$

$\frac{V_b + 12}{4k\Omega} - \frac{V_d}{4k\Omega} = 1mA$

$V_b + 12 - V_d = 4$

$V_b - V_d = -8$

Problem 4.3-3

Determine the values of the power supplied by each of the sources in the circuit shown in Figure P 4.3-3.
Note: The voltages of the other nodes with unknown voltage can be expressed in terms of the node voltage v_a in Figure P 4.3-3.

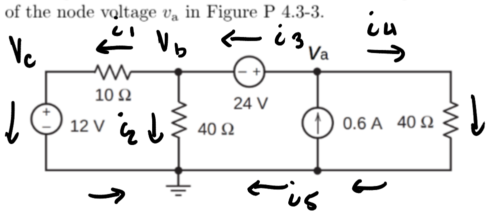

Figure P 4.3-3

$V_a - V_b = 24$

$V_c = 12$

$\frac{V_b - V_c}{10} = i_1$

$\frac{V_a}{40} = i_4$

$\frac{V_b}{40} = i_2$

$0.6 = i_1 + i_2 + i_4$

$i_4 = i_5 + 0.6$

$i_1 + i_2 = i_3$

$i_1 + i_2 + i_5 = 0$